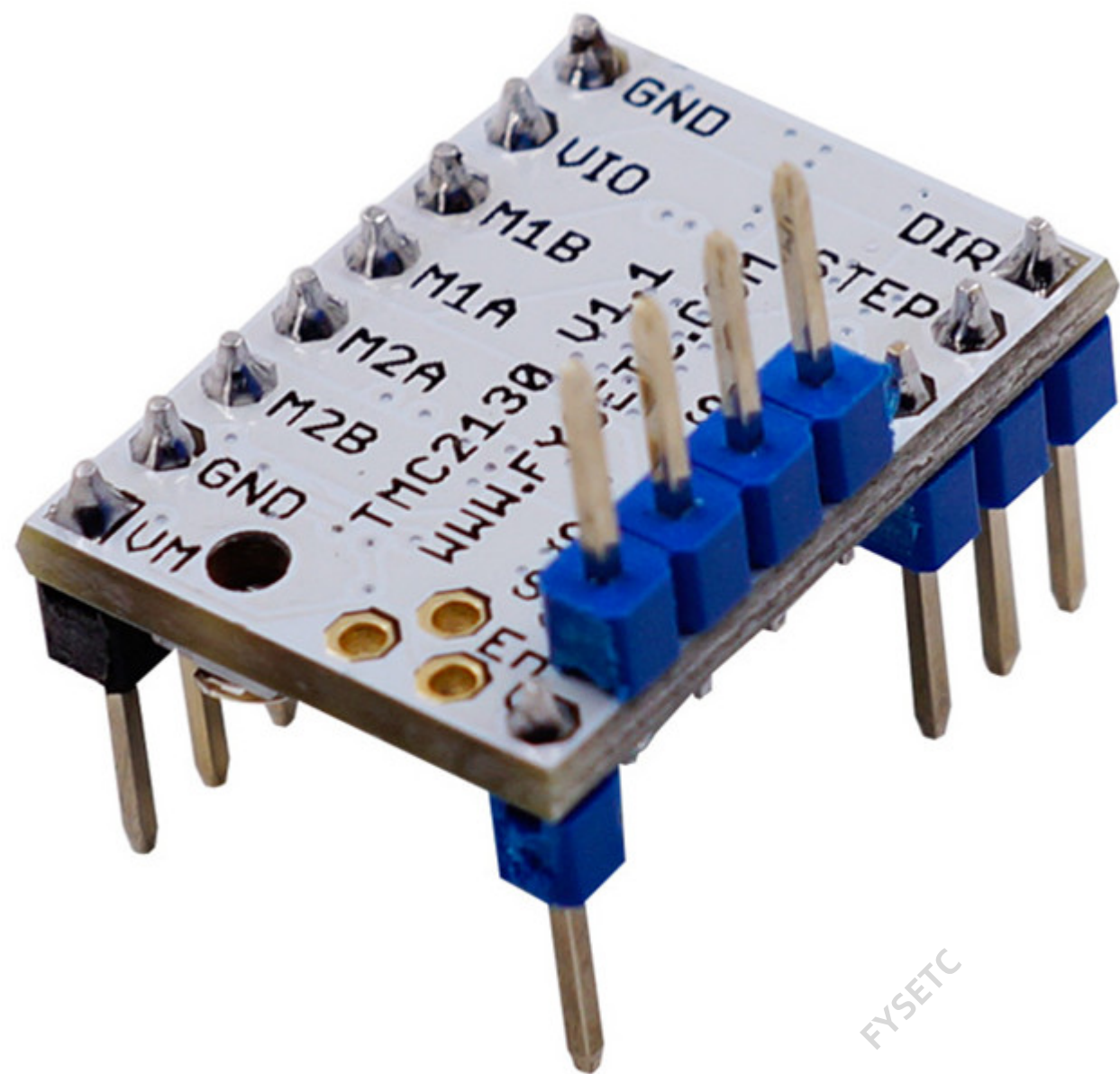


TMC2130

Product Introduction



The TMC2130 provides an integrated motor driver solution for 3D-Printing, Cameras, Scanners and other automated equipment applications. The device has an integrated microstepping indexer, the sensorless stall detection technology StallGuard2™, the sensorless load dependent current control CoolStep™ and the completely noiseless current control mode StealthChop™ and is intended to drive a bipolar stepper motor.

Versions Difference

Versions	difference
V1.0	The default is standalone mode, If you need SPI mode, you must config by yourself.
V1.1	SPI mode, SPI pin out (top) , you must connect wires to your board, and config the firmware for it, otherwise the stepper can't run.
V1.2	SPI mode, SPI&DIAG pin out (bottom) , it's used for F6.

Features

- SPI configuration interface (up to 4MHz)
- Up to 256 native microsteps (without interpolation)
- Reduce energy consumption by 75%
- High accuracy sensorless motor load detection
- High load driver will not lost step
- Automatic stealthChop and spreadCycle switchover depending on velocity
- Hardware compatible with StepStick and Pololu A4988 Stepper Driver
- Components on bottom PCB side for better heat emission
- Automatic standby current reduction
- stealthChop - for quiet operation and smooth motion
- spreadCycle - highly dynamic motor control chopper

Technical Specifications

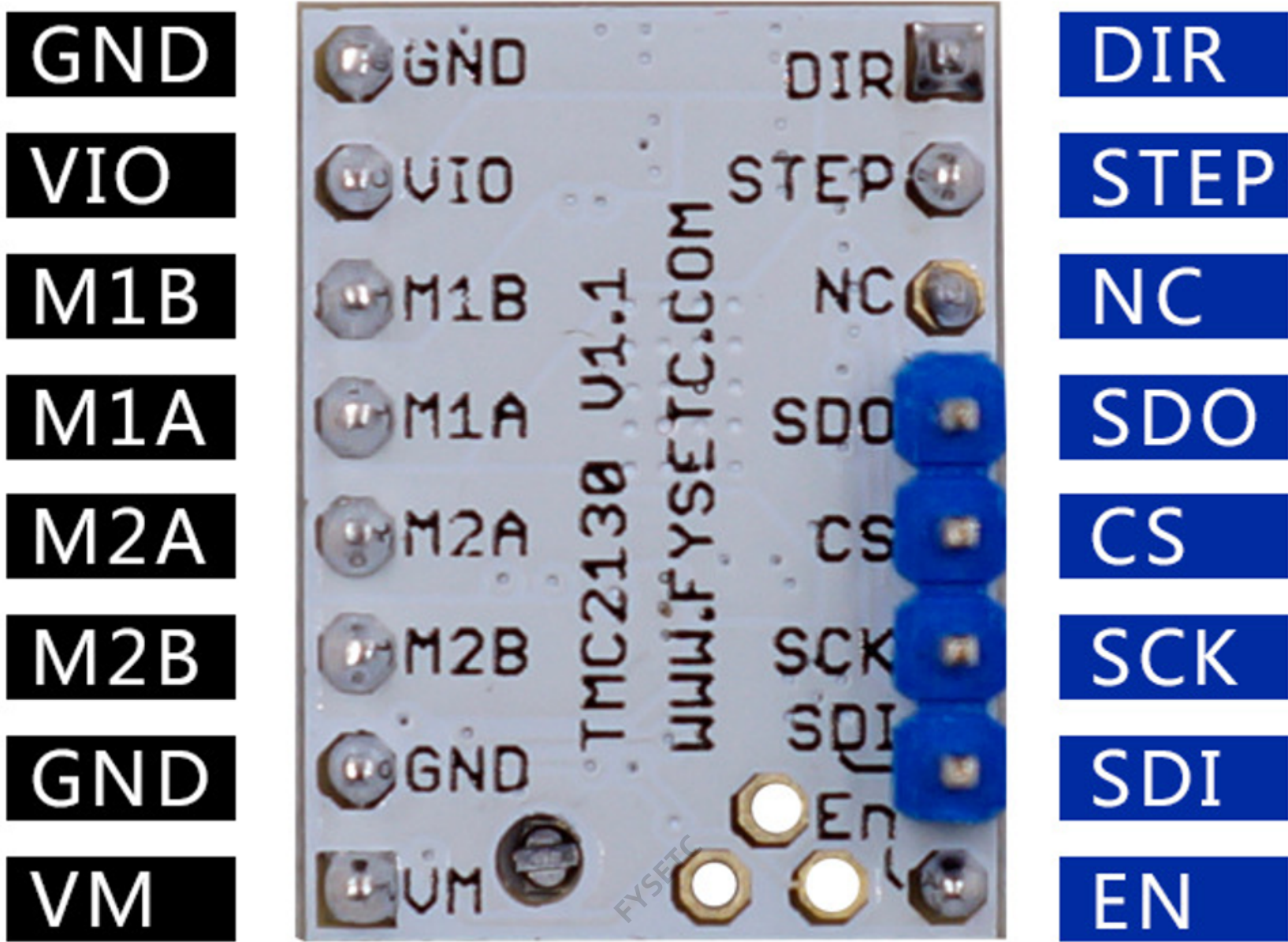
Model	TMC2130
Interface	Step/Dir or SPI
Configuration	CFG Pins or SPI
Native Microsteps	up to 1/256
microPlyer Microsteps	1/256
Logic Voltage (VIO)	3-5V
Motor Voltage (VM)	5.5-46V
Motor Phase Current max	1.2A RMS, 2.5A Peak
Internal V- Regulator	enabled
RDSon	>=0.5 Ohm
stealthChop (quiet)	yes
spreadCycle	yes
coolStep	yes
stallGuard	yes
dcStep	yes

Pin Functions

FYSETC

TMC2130

Motor Drive Pin Descriptions



Pin	Function
Power Supply	
GND	Ground
VM	Motor Supply Voltage
VIO	Logic Supply Voltage
Motor Outputs	
M1A	Motor Coil 1
M1B	Motor Coil 1
M2A	Motor Coil 2
M2B	Motor Coil 2
Control Inputs	
STEP	Step-Signal Input
DIR	Direction-Signal Input
TMC2130 (SPI_MODE=1)	
EN	Enable Motor Outputs: GND=on, VIO=off, OPEN=Auto-Power-Down
SDO/CFG0	Serial Data Output (SPI)
SDI/CFG1	Serial Data Input (SPI)
SCK/CFG2	Serial Clock Input (SPI)
CS/CFG3	Chip Select Input (SPI)

Pin	Function
DCEN/CFG4	dcStep Enable Input (GND=off)
DCIN/CFG5	dcStep Gating Input
DIAG0	Diagnostics Output 0
DIAG1	Diagnostics Output 1

Interface Resources

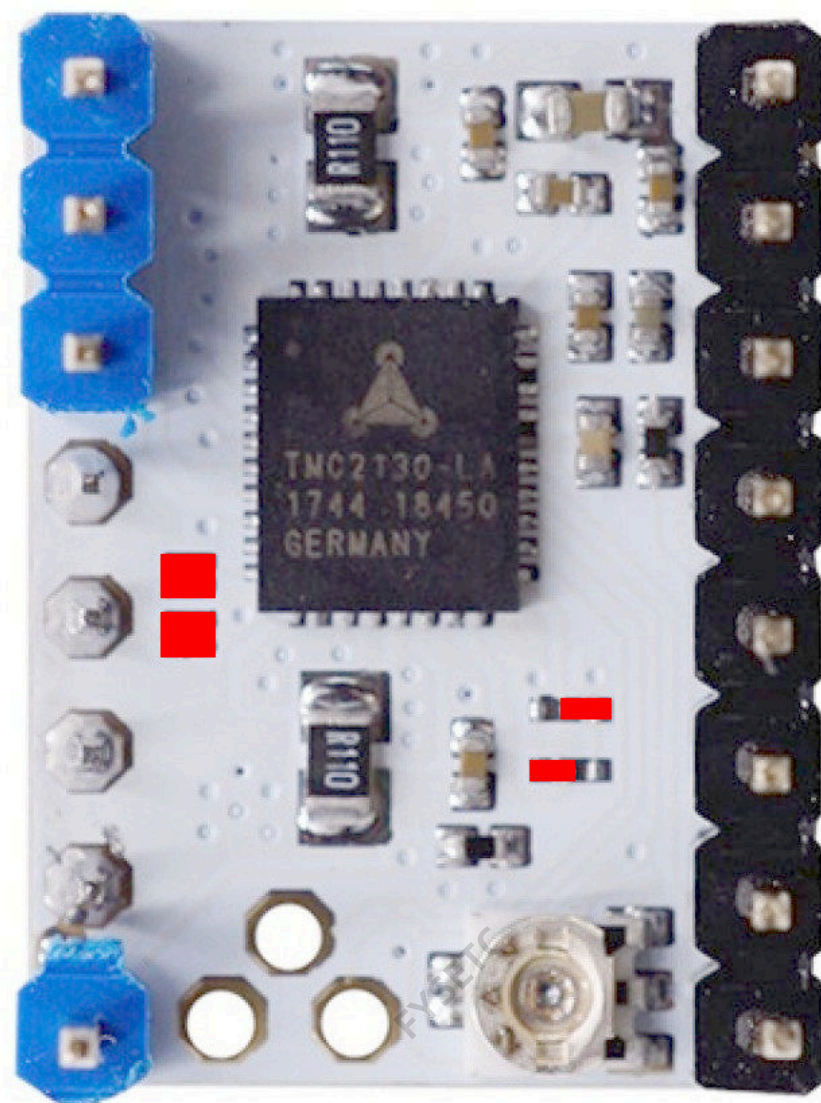
SPI Mode

v1.0

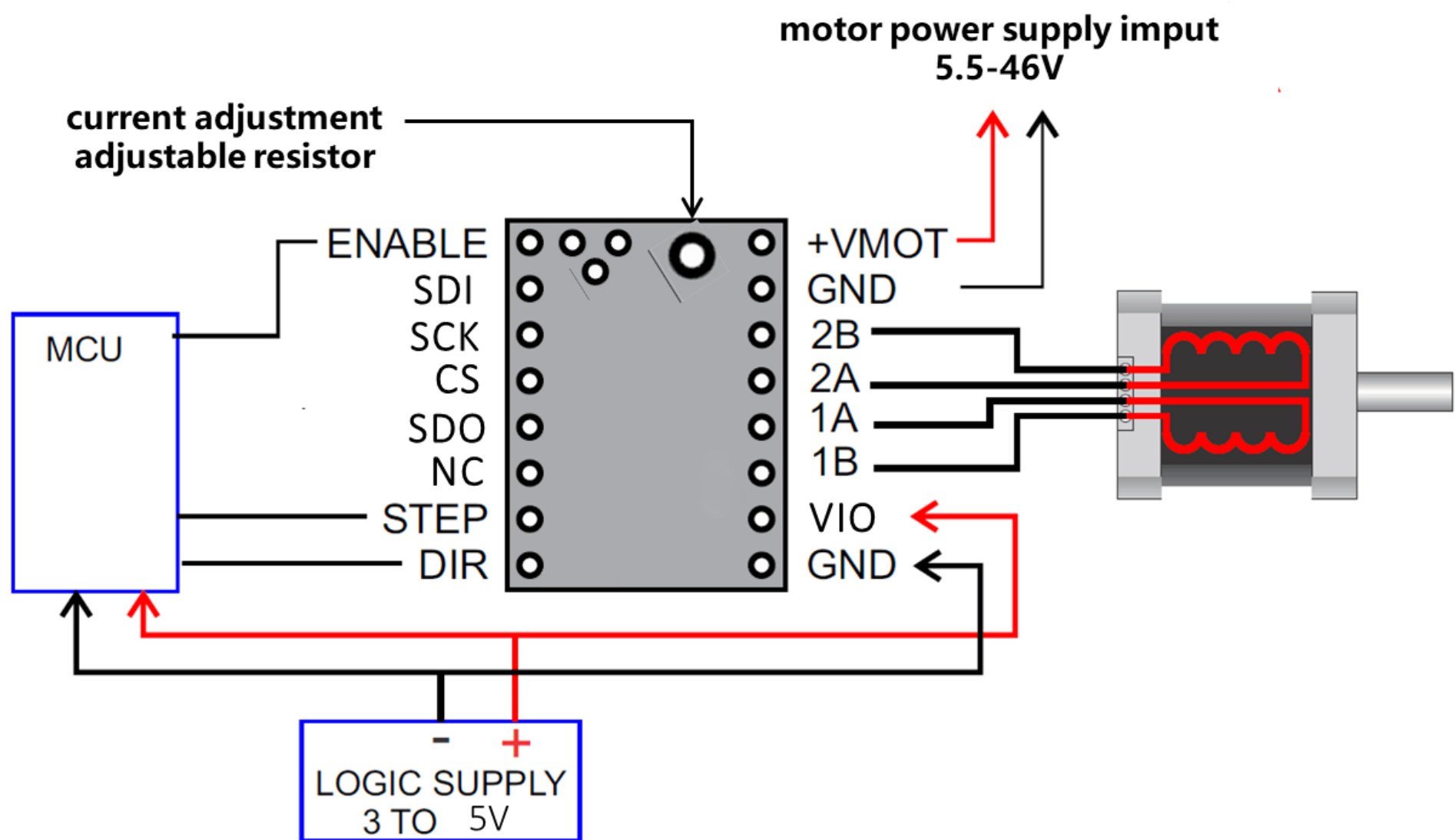
The FYSETC v1.0 drivers come pre-configured in standalone mode. To operate in SPI mode you need to modify three solder bridges on the driver PCB.

Some versions of the FYSETC v1.0 drivers come with a solder bridge left of the chip, some come with a bridging resistor. This connection needs to be opened for SPI connection to work. The two smaller bridges need to be configured as shown.

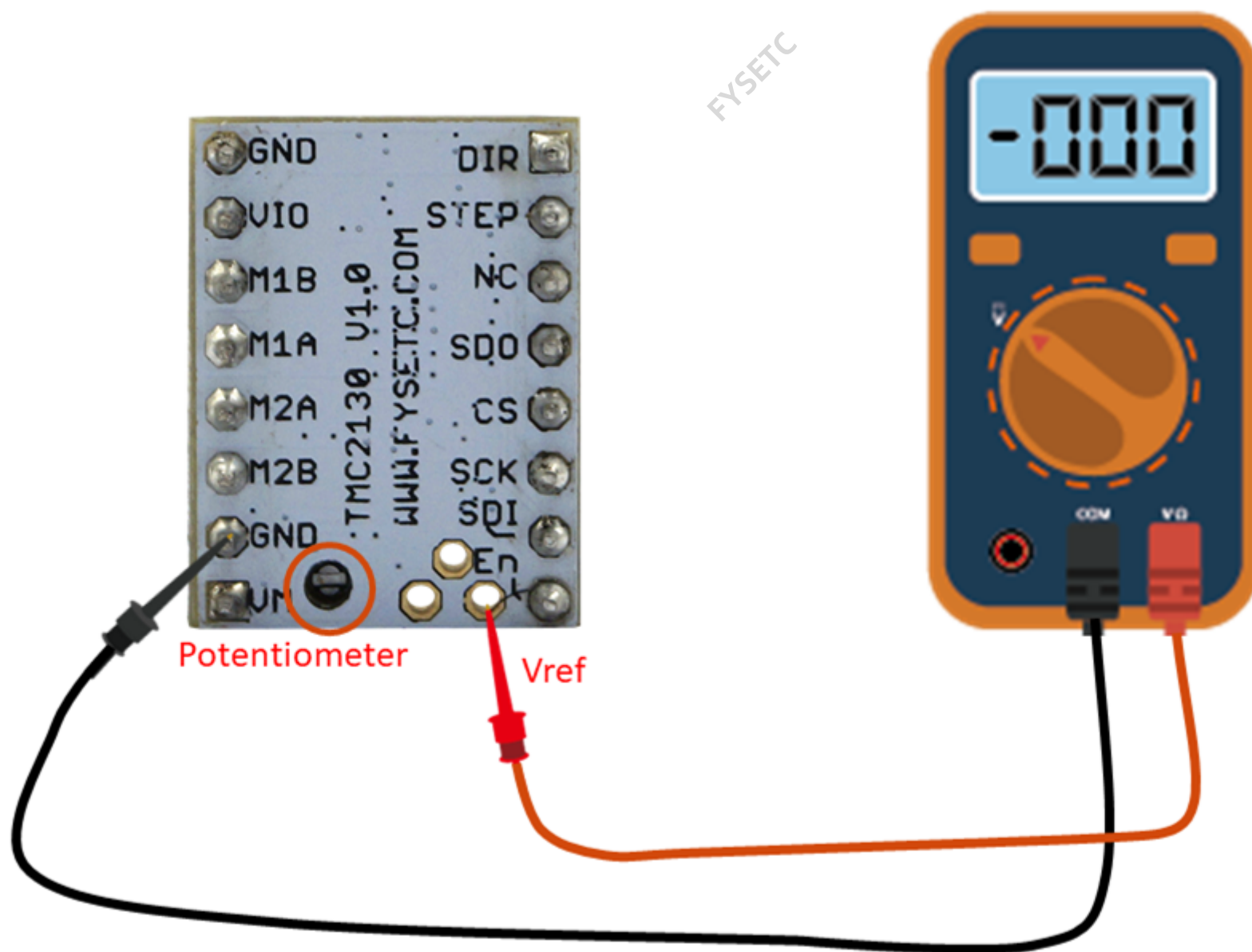
FYSETC



Wiring Diagram



Motor Current Setting



The best way to set the motor current is by measuring the voltage on the Vref pin (0...2.5V) and adjusting the voltage with the potentiometer. The maximum settable motor current is 1.77A RMS (0.110hm sense resistors), but the SilentStepSticks can only be used up to 1.2A RMS.

$$I_{rms} = (V_{ref} * 1.77A) / 2.5V = V_{ref} * 0.71$$

$$V_{ref} = (I_{rms} * 2.5V) / 1.77A = I_{rms} * 1.41 = I_{max}$$

Vref -> Voltage on Vref pin

I_{rms} -> RMS (Root Mean Square) current per phase (I_{rms} = I_{max} / 1.41)

I_{max} -> Maximum current per phase (I_{max} = I_{rms} * 1.41)

!!! note □ 1.V_{ref} measures G_{nd} and the voltage at the middle of the potentiometer.

□ 2.Do not connect the motor when measuring the voltage, otherwise it is easy to burn the driver.

□ 3.Power should be connected when measuring voltage, do not just connect USB power supply.

□ 4.**Please pay special attention to directions!**

!!!

RMS to Voltage Calculator

Reference Voltage (V) = RMS Current (A) × 1.41

Shop

- [TMC2130](#)

Tech Support

Please join our Discord server - <https://discord.gg/S9nZUCnDgm>

FYSETC